BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

DIGITAL CIRCUITS (2023 (SPECIAL))

Exam Date: Tuesday, 3 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2023 (Special)	Digital Electronics	100403	Special Examination - 2023

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

Special Examination - 2023

DIGITAL ELECTRONICS

Time: 03 Hours | Code: 100403 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven question only: **[14]**

1. Define Dynamic RAM.
2. The resolution of a 10-bit AD converter for an input range of 10 V is approximately _____
3. The parameter through which 16 distinct values can be represented is known as _____
4. Define Multiplexing.
5. Write the significance of Truth table.
6. Write the differences between Combinational and Sequential circuit.
7. A hexadecimal odometer displays F52F. The next reading will be _____
8. The basic property of Sequential circuit is _____
9. How many entries will be in the truth table of a 4-input NAND gate?
10. What is Dual Slope ADC?

Q.2 Solve this question : **[14]**

1. What are weighted, non-weighted, cyclic and self-complementary codes? Explain each with examples.

Q.3 Solve this question : **[14]**

1. Design a BCD to 7-segment display decoder circuit using logic gates.

Q.4 Solve this question : **[14]**

1. Design a 3-bit parallel comparator A/D convertor that provides output in 2's complement format.

Q.5 Solve both questions : **[14]**

1. Design full adder using 8:1 mux.

2. Design full adder using 4:1 mux.

Q.6 Solve this question : **[14]**

1. What is binary shift register? Write down their application.

Q.7 Solve this question : **[14]**

1. Minimize the following expression using k-map
   $f(P, Q, R, S) = \Sigma m(0, 1, 4, 5, 7, 12, 13)$

Q.8 Solve this question : **[14]**

1. Describe the procedure to design Mod-6 counter.

Q.9 Write short notes on any two of the following: **[14]**

1. i. RAM
   ii. ROM
   iii. CMOS logic
   iv. Operation of TTL logic circuit working as NAND Gate